

TW16: Handling Time-Series Data

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Create a reliable time index, derive lagged and rolling measures, compare growth rates, change frequency, and validate without future leakage.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Audit and regularise a time series

Scenario

Monthly sales must be checked before any trend calculation.

INSTRUCTIONS

1. Import and parse month, sort it, and create a DatetimeIndex.
2. Verify monotonic order, uniqueness, start/end, and inferred frequency.
3. Use `asfreq('MS')` to expose absent periods.
4. Distinguish completely absent months from missing sales values.

REQUIRED SUBMISSION

- A regular monthly table.
- A complete time-series audit record.
- Assertions for order, uniqueness, and expected frequency.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Create lags, differences, and rolling summaries

Scenario

The manager wants short-term movement and smoothed trend without using future information.

INSTRUCTIONS

1. Create 1-month, 3-month, and 12-month lags.
2. Calculate first and 12-month differences.
3. Calculate 3-, 6-, and 12-month trailing means and a 6-month standard deviation.
4. Plot raw sales with all rolling means.
5. Explain missing initial values and smoothing lag.

REQUIRED SUBMISSION

- A lag-and-rolling feature table.
- A chronological multi-line chart.
- A correct leakage and lag interpretation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Compare growth measures

Scenario

The board uses month-on-month, year-on-year, and CAGR interchangeably.

INSTRUCTIONS

1. Calculate MoM and YoY growth for sales and orders using `fill_method=None`.
2. Write a validated CAGR function and apply it to sales using elapsed years.
3. Report the latest valid comparison periods explicitly.
4. Explain what each measure answers and what CAGR hides.

REQUIRED SUBMISSION

- A growth table and reusable CAGR function.
- Latest-period briefing values.
- A concise comparison of the three growth concepts.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Change frequency and test missing-value assumptions

Scenario

Quarterly reporting is required, but three monthly sales values are missing.

INSTRUCTIONS

1. Compare time interpolation, linear interpolation, and trailing-mean estimates only at missing months.
2. Preserve a missingness indicator.
3. Create quarterly totals for sales/orders and quarterly mean marketing spend.
4. Rebase sales, orders, and marketing to 100 at the first valid observation.
5. Explain the difference between downsampling, upsampling, and imputation.

REQUIRED SUBMISSION

- An interpolation sensitivity table.
- Quarterly and rebased tables with validation checks.
- One frequency-change chart and a documented preferred assumption.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Design time-aware validation and a briefing

Scenario

A forecasting prototype must avoid random splitting and report its evaluation periods.

INSTRUCTIONS

1. Reserve the final 12 months as a chronological test set.
2. Create four expanding TimeSeriesSplit validation folds from earlier data.
3. Assert that every validation period follows its training period.
4. Summarise seasonal monthly patterns and recent growth.
5. Create a business briefing with assumptions, dates, missing treatment, and next modelling step.

REQUIRED SUBMISSION

- A final holdout and fold-window table.
- Seasonal-profile and latest-growth evidence.
- A 150-to-200-word time-series briefing.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____